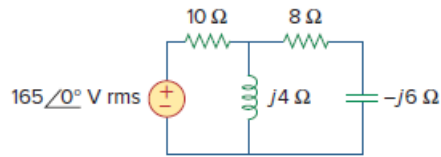


Problem

Calculate the power factor of the entire circuit of Fig. 11.19 as seen by the source. What is the average power supplied by the source?

Figure 11.19:



Step-by-step solution

Step 1 of 4

Consider the following circuit diagram:

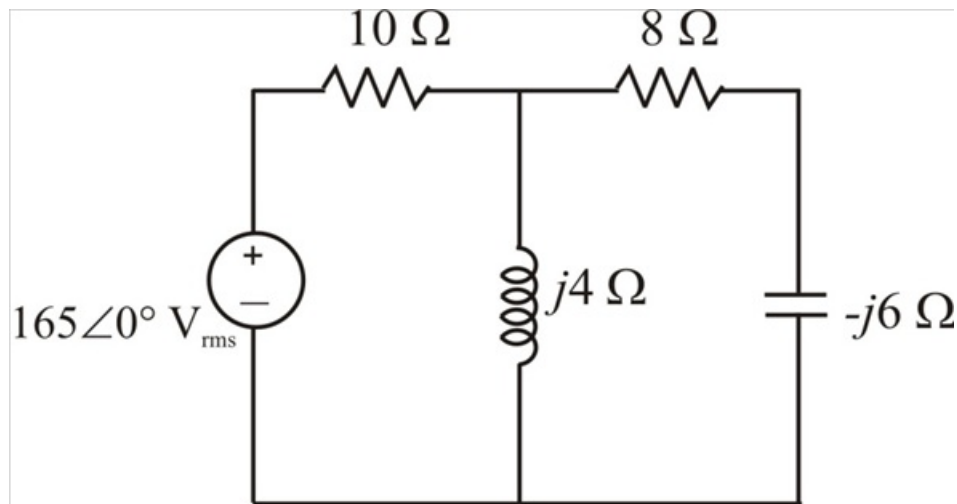


Figure 1

Step 2 of 4

Calculate the value of impedance $\mathbf{Z}$ from the circuit in Figure 1:

$$\begin{aligned}\mathbf{Z} &= 10\ \Omega + \frac{(j4\ \Omega)(8\ \Omega - j6\ \Omega)}{(j4 + 8 - j6)\ \Omega} \\ &= 10 + \frac{(j4)(8 - j6)}{j4 + 8 - j6} \\ &= (11.882 + j4.470)\ \Omega \\ &= 12.695\angle 20.618^\circ\ \Omega\end{aligned}$$

Step 3 of 4

The following is the power factor:

$$\begin{aligned}\text{pf} &= \cos(20.618^\circ) \\ &= 0.9359 \\ &= 0.9359 \text{ lagging}\end{aligned}$$

The power factor is lagging because the current lags the voltage.

Therefore, the required power factor is 0.9359 lagging.

Step 4 of 4

Calculate the rms value of current $\mathbf{I_{rms}}$:

$$\begin{aligned}\mathbf{I_{rms}} &= \frac{\mathbf{V_{rms}}}{\mathbf{Z}} \\ &= \frac{165\angle 0^\circ \text{ V}}{12.695\angle 20.618^\circ \Omega} \\ &= \frac{165\angle 0^\circ}{12.695\angle 20.618^\circ} \\ &= 12.997\angle -20.618^\circ \text{ A}\end{aligned}$$

Calculate the average power $\mathbf{P}$ supplied by the source:

$$\begin{aligned}P &= (\mathbf{I_{rms}} \mathbf{V_{rms}})(\text{pf}) \\ &= (12.997 \text{ A})(165 \text{ V})(0.9359) \\ &= (12.997)(165)(0.9359) \\ &= 2.007 \text{ kW}\end{aligned}$$

Therefore, the value of average power $\mathbf{P}$ supplied by the source is 2.007 kW.